

Randall Sim

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Education

Brown University

B.S. in Computer Science, GPA: 4.0

Providence, RI

Sep 2025 — May 2028

Stony Brook University

Computer Science, GPA: 4.0 (Transferred to Brown)

Stony Brook, NY

Aug 2024 — May 2025

Skills

Languages: Javascript/Typescript, Java, Python, Golang, Rust, C#

Frameworks: React, Next.js, Spring Boot, Express, Flask, .NET

Tools: AWS, Docker, Git, Linux, PostgreSQL, Redis, RabbitMQ

Work Experience

Microsoft

Redmond, WA

Software Engineer Intern

May 2026 — Aug 2026

- Shipped an AI feature for Microsoft Copilot in Intune that converts natural-language device queries into interactive data visualizations, **deployed to Microsoft's internal dogfood environment** and validated against Microsoft IT's 2M+ device fleet.
- Removed a **50K-device rendering limit** by rewriting KQL queries to aggregate server-side, enabling visualizations over unlimited-size datasets.
- Developed the full stack: a React graph-editing interface and a C# GPT skill on Microsoft's Security Copilot platform, integrated into the existing Intune portal.
- Presented the feature in enterprise customer calls; partnered with the Custom Reports team to scope cross-feature export of graph data.

Stony Brook Campus Residences

Stony Brook, NY

Software Engineer Intern

Sep 2024 — May 2025

- **Rebuilt PrintStats**, Stony Brook's official printer dashboard. Migrated the entire C# .NET backend to Express.js, utilizing asynchronous functions to achieve a **12x improvement (6 min → 30 sec)** in loading printer metadata via SNMP.
- Built and maintained dashboards (C# .NET, Express, React, SQL) used by 100+ professional staff to monitor campus Scalas (TVs), computer closets, and KACE-managed endpoints.
- Modernized CI/CD by replacing manual Remote Desktop + IIS setup with Github Actions and Windows dev containers.

Personal Projects

epsilon.stuysu.org

- Founded Epsilon, an all-in-one platform serving **3,300+ daily users** and **400+ student organizations** at Stuyvesant HS—**sole engineer for first year**, now maintained by the school.
- Designed a PostgreSQL row-level security model enforcing role-based permissions (member, admin, and org-level access) across **100+ row level security policies and 30+ schema migrations**; integrated Google Calendar and OAuth via **20+ Deno edge functions**.
- Deployed a responsive frontend on Netlify using React, Typescript, and react-router.

internships.dev

- Co-founded internships.dev, a job-alerts platform tracking 2000+ tech internships and 200+ job boards, sending users live job alerts (**~1 hr → a few days**) faster than competitors.
- Architected a distributed web scraper in Python, utilizing **Redis** and **Tailscale** to coordinate execution among **10+ nodes**.
- Created an alerts microservice with **RabbitMQ** and **Node.js** to deliver notifications to Discord, Github, and email.
- Developed backend with **Spring Boot**, implementing **OAuth exchange code protocol** using Redis.
- Established CI/CD pipelines with Github Actions to efficiently update 4 microservices.

[Semantic Fetch](#)

- Built a **Rust** AI agent (rig-rs) and **TypeScript SDK** that route natural-language queries to REST endpoints, with a client-side cache reducing repeated-query latency to match direct API calls.
- Demonstrated the SDK in a full-stack reference app (**Go** REST backend, **Next.js** frontend) showing natural-language routing integrated into a real CRUD application.

Relevant Coursework

Data Structures & Algorithms, Programming Abstractions, Systems Programming, Analysis of Algorithms, Fundamentals of AI, Software Engineering, Computer Networking, Applied Cryptography, Discrete Math, Calculus I, II, and III.